

Homework 1

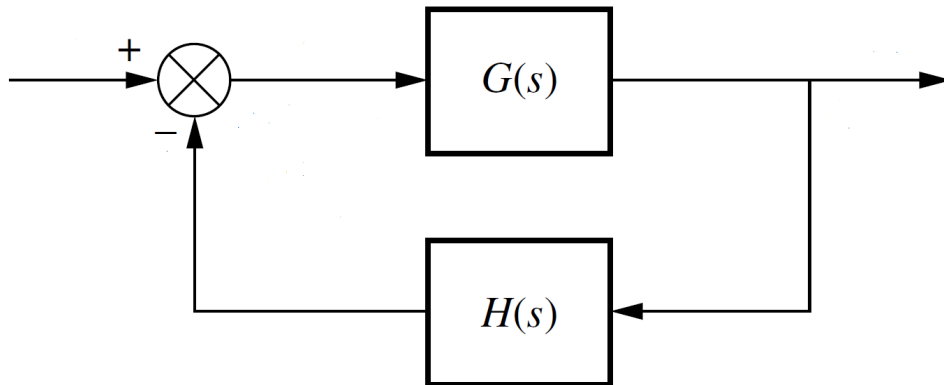


Figure 1: Control System

1. Consider the control system in Figure 1 with

$$G(s) = \frac{K_1}{s(s+1)} \quad H(s) = 1 + K_2s$$

Find the value of K_1 and K_2 such that the peak overshoot is 0.2 and the peak time is 1 second.

Answer: $K_1 = 12.3$ and $K_2 = 0.175$

2. Consider the control system in Figure 1 with

$$G(s) = \frac{K}{(s^2 + 8s)} \quad H(s) = 1$$

Find the value of K such that the peak overshoot is 0.16.

Answer: $K = 64$

3. Consider a second order system with the rise time 0.168 sec and the peak time 0.25 sec. Find the poles of the system.

Answer: $s = -7.58 \pm i12.56$

4. For the following transfer functions, find the damping ratio, natural frequency and the nature (over-damped, underdamped, undamped) of the response

- a) $G(s) = \frac{400}{s^2 + 12s + 400}$
 b) $G(s) = \frac{900}{s^2 + 90s + 900}$
 c) $G(s) = \frac{225}{s^2 + 30s + 225}$
 d) $G(s) = \frac{625}{s^2 + 625}$

Answer:

- a) $\zeta = 0.3, \omega_n = 20$: System is Underdamped
 b) $\zeta = 1.5, \omega_n = 30$: System is Overdamped
 c) $\zeta = 1, \omega_n = 15$: System is Critically damped
 d) $\zeta = 0, \omega_n = 25$: System is Undamped

5. The unit step response of an under-damped second order system has the steady-state value of -2 . Which one of the following transfer functions has this property?

- (a) $\frac{-2.24}{s^2 + 2.59s + 1.12}$
 (b) $\frac{-3.82}{s^2 + 1.91s + 1.91}$
 (c) $\frac{-2.24}{s^2 - 2.59s + 1.12}$
 (d) $\frac{-3.82}{s^2 - 1.91s + 1.91}$

6. Consider the control system in Figure 1 with

$$G(s) = \frac{14.4}{s(1 + 0.1s)} \quad H(s) = 1$$

Find the damped frequency of oscillations in the unit step response of this system.

Answer: $\omega_d = 10.90 \text{ rad/s}$

7. Consider the polynomial $D(s) = s^5 + s^4 + 2s^3 + 2s^2 + 3s + 15$. Find the number of roots of $D(s)$ which lie in the RHP of the complex plane.

Answer: 2

8. Consider the transfer function $T(s) = \frac{10}{s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3}$. Find the number of poles of $T(s)$ in the RHP of the complex plane.

Answer: 2

9. Consider the transfer function

$$T(s) = \frac{ps^2 + 3ps - 2}{s^2 + (3 + p)s + (2 - p)}$$

where p is a positive real parameter. Find the range of p for which $T(s)$ is BIBO stable.

Answer: $p < 2$

10. Consider the polynomial $D(s) = s^8 + s^7 + 12s^6 + 22s^5 + 39s^4 + 59s^3 + 48s^2 + 38s + 20$. Find the number of roots of $D(s)$ in the LHP, RHP and the imaginary axis of the complex plane.

Answer: 2 roots in LHP, 2 roots in RHP and 4 roots on imaginary axis